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RESEARCH ARTICLE

Convolution Self-Guided Transformer for Diagnosis and Recognition of Crop Disease in Different Environments

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ABSTRACT Accurately diagnosing crop diseases is crucial for agricultural productivity and food safety. This study addresses the challenge by developing an AI crop disease diagnosis platform, leveraging the strengths of Convolution Neural Networks (CNNs) and Vision Transformers. The proposed Convolution Self-Guided Transformer (CSGT) model integrates CNN's local feature extraction with the Self-Guided Transformer's (SGT) global information processing, enhancing the precision and efficiency of agricultural diagnostics. We selected two datasets with characteristics from Jiangsu and Guangdong provinces to validate our model, respectively representing controlled and uncontrolled agricultural environments. The CSGT model demonstrates exceptional performance in diagnosing crop diseases across diverse backgrounds, overcoming the limitations of current models, especially in complex settings. The CSGT model's CNN layer, hybrid-scale, and self-guided attention mechanisms ensure accurate diagnoses, even in the presence of background clutter. CSGT has an accuracy of 96.9%(Apple), 95.8%(Corn), 96.1%(Grape), and 96.5%(Tomato) when classifying crop diseases in a stable environment, and 95.8%(Rice) when classifying diseases in a complex environment. The research results are anticipated to enhance the effectiveness and stability of natural crop disease recognition applications.

INDEX TERMS Vision transformer, crop protection, crop disease diagnosis, self-guided attention, AI for agriculture.

I. INTRODUCTION

Diagnosing crop diseases has long been a significant concern for countries worldwide. Recognizing these crop diseases holds immense importance as they directly impact the safety of various aspects of the agricultural industry and food supply chain [1]. Efficient recognition of crop diseases is paramount for China, as it directly impacts crop yield and agricultural development, posing a significant threat. To promote the development of new quality productive forces and align

with the “AI+” policy, our plan involves leveraging an AI diagnosis platform to expedite the process of diagnosing crop diseases. By embracing this approach, we aim to improve agricultural production efficiency. China's agricultural sector encompasses vast land areas, spans diverse geographical regions, and involves a significant agricultural population. Nevertheless, China's notable challenge is the shortage of agricultural experts [2]. Our strategy integrates high-quality agricultural research resources within the Guangdong-Hong Kong-Macao Greater Bay Area. Additionally, we aim to collaborate with experts from agricultural universities to establish an AI crop disease diagnosis platform. This platform

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will be specifically designed to cater to the needs of grassroots agricultural technicians and farmers nationwide.

Rapid AI progress has prompted diverse industries to use AI technology to solve challenges in their domains in recent years. AI offers potent methodologies for addressing challenges in various domains, driving advancements, and notably progressing in crop disease object recognition [3] and classification tasks [4]. When a disease affects a crop, it typically exhibits symptoms on its leaves, stems, flowers, or fruits [5]. The pathological characteristics on the leaves are usually visible and easily identifiable. As a result, determining the specific type of crop disease primarily relies on recognizing the pathological features present in the leaves. The existing literature clearly indicates that numerous researchers have integrated deep-learning techniques in the field of agriculture. Moussafir et al. [6] developed a hybrid model with genetic algorithms and transfer learning, achieving 98.1% accuracy on tomato leaf disease recognition. Latif et al. [7] proposed an improved CNN model for rice plant disease detection based on a modified VGG19 architecture, which yielded the highest accuracy of 96.08% using a non-normalized augmented dataset. Guo et al. [8] presented the Convolution Swin Transformer, which excelled in detecting plant diseases with an accuracy of 98.2% in categorization and maintained 79.5% accuracy under 30% salt noise, showcasing its robustness. Feng et al. [9] designed the lightweight FoldNet, which employs isotropic CNN and achieved a 99.84% recognition rate on the PlantVillage dataset and 90.49% on the FGVC8 dataset with a minimal parameter count of less than 1M, indicating its efficiency. Chen et al. [10] introduced a stacking ensemble model, Es-MbNet, which integrated three lightweight CNNs and attained an average accuracy of 99.37% on local datasets and 99.61% on the PlantVillage dataset. Xu et al. [11] enhance potato disease severity classification with a two-stage model featuring Coformer and SegCoformer, achieving 97.86% accuracy and 88.50% mIoU in segmentation, streamlining grading processes. Sarkar et al. [12] introduce CDC-Deconv, a deep neural network utilizing network deconvolution and AReLU, which excels with 100% accuracy on the Grape dataset, highlighting deep learning's role in precise crop disease identification. Gautam et al. [13] developed ESDNN for mango leaf disease detection, with a 98.57% accuracy rate, demonstrating the efficacy of ensemble deep learning models combined with machine learning for disease identification. Wang et al. [14] propose a CNN-enhanced method with DSGC and AFFM, improving computational efficiency and feature learning for crop disease detection, attaining high classification metrics. Xu and Li [15] present FRseNet, a deep residual network with feature recalibration, for recognizing a range of crop diseases, emphasizing the impact of advanced neural networks in agricultural diagnostics. Agricultural images with complex backgrounds refer to crop images containing diverse scenes and distracting elements like weeds, shadows, and soil. The previous studies did not use agricultural image data with complex backgrounds, nor

did they perfectly combine CNN and Transformer to apply to complex scenes of agricultural production. Crop disease detection in complex scenarios plays a vital role in food security.

Given that the construction of this platform requires significant resources and involves data sources from various regions, we plan to initially select specific locations as pilot sites to conduct research. Our initial research will use two datasets with characteristics of major local agricultural provinces: (1) **DataSet1**, the Nanjing agricultural pest and disease dataset from Jiangsu Province [16] and (2) **DataSet2**, the Rice leaf image dataset with characteristic crops in Guangdong Province [17]. In the field of agriculture, data acquisition is often limited by factors such as safety, ethics, resource availability, and cost. As a result, researchers frequently face challenges in obtaining many samples, leading to the construction of relatively small-scale image datasets. The data utilized in this study also possesses these characteristics, representing a small sample dataset.

Two distinct datasets have been chosen for this study, each with different background content. **DataSet1** comprises controlled, stable, and undisturbed agricultural disease leaf images, providing a consistent background. In contrast, **DataSet2** features an uncontrolled and complex background, including planting scenes and images of agricultural diseased leaves that may contain various disturbances. Image data with different backgrounds are selected to be compatible with the data forms reported by different people. As depicted in Fig. 1, grassroots agricultural technicians usually collect diseased crop leaves directly from the planting site and bring them back to the office environment. This approach enables them to utilize controlled conditions that minimize background interference, ensuring clear photographs can be captured. Subsequently, these images are uploaded to the analysis platform for further evaluation and analysis. On the other hand, ordinary farmers can use their mobile devices, such as smartphones, to capture photographs of crop leaves directly in the field. These images may contain background interference due to the uncontrolled environment. These images are then uploaded to the same platform for diagnostic analysis.

The images captured display the affected areas of the crop leaves, which can vary in the extent of the disease, impacting either a portion or the entire leaf. CNNs are particularly effective in such scenarios due to their proficiency in capturing local image features, crucial for identifying diseased areas on crop leaves. With their automatic feature extraction capabilities, CNNs simplify the model development process by eliminating the need for manual feature engineering. This ability, coupled with their enhanced stability and accuracy across various environments, makes CNNs ideal for precision tasks such as agricultural image analysis. However, despite their strengths, CNNs have inherent limitations. The restricted field of view, dictated by the kernel size, results in a significant proportion of global information being lost, which can hinder comprehensive analysis.

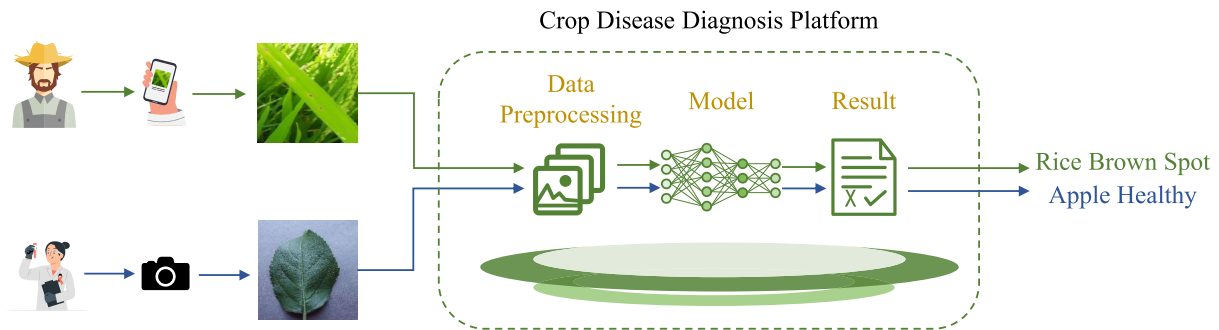


FIGURE 1. This crop disease diagnosis platform allows ordinary farmers and grassroots agricultural technicians to upload photos for quick crop health diagnostics.

To address the limitations of CNNs, Vision Transformers (ViTs) have emerged as a complementary approach, leveraging a self-attention mechanism to gather both local and global information from images. Unlike CNNs, ViTs allow the model to consider the entire image context right from the start of training, which can be advantageous in capturing broader patterns and relationships. However, this global perspective comes with trade-offs. While ViTs excel at processing holistic image information, they may not capture localized features as effectively as CNNs, which are critical in diagnosing specific crop diseases. Moreover, the higher computational requirements of Transformers can be a challenge in resource-constrained environments, such as agriculture. The standard attention mechanism in Transformers also tends to distribute focus evenly across the image, potentially overlooking critical areas like crop lesions, which could slow down the diagnostic process. Thus, a combined approach that leverages the strengths of both CNNs and ViTs might offer a more robust solution for agricultural image analysis.

In this study, we maintain the SG-Former as our base and explore the benefits of combining CNNs with Vision Transformers (ViT) in crop disease recognition and classification. The Convolution Self-Guided Transformer (CSGT) integrates the local feature extraction advantage of CNNs with the global feature superiority of Transformers, aiming to bolster the accuracy of crop disease diagnostics. Its hybrid-scale and self-guided attention mechanism allows the model to adeptly process images with diverse backgrounds and efficiently extract disease features from controlled or field settings. CSGT establishes an algorithmic foundation for AI agriculture platform development that is adaptable to various agricultural contexts. This novel integration is designed to overcome the limitations of current models, especially in complex backgrounds, potentially elevating the precision and efficiency of agricultural diagnostics and offering significant support to grassroots agricultural technicians and farmers nationwide. The main contributions of this study are the following:

(1) A new model, CSGT, is proposed to diagnose crop diseases specifically for different user groups and different data backgrounds of the AI crop disease diagnosis platform. It significantly improves the accuracy of crop

disease recognition under stable and complex background conditions.

(2) Equipped with hybrid-scale and self-guided attention mechanisms, the CSGT model effectively recognizes various types of crop diseases within both stable and complex environments, guaranteeing precise diagnoses. The integration of CNN Layers allows for the efficient extraction of local features, even in the presence of background clutter, which is crucial for accurately recognizing disease indicators.

(3) We have contributed to the field of agricultural technology by providing a detailed analysis and experimental validation of the CSGT model on two crop disease datasets. Experimental results show that the CSGT model surpasses previous studies' proposed methods for crop disease classification in both stable and complex environments.

II. RELATED WORK

A. CNN IN CROP DISEASE DIAGNOSIS

In the field of crop disease recognition, CNN's development has ushered in innovative models. Rakib et al. [18] introduced a lightweight quantized CNN (Q-CNN) approach for recognizing nine different plant diseases using the ESP32-CAM IoT device. Fan and Guan [19] presented VGNet, a modified VGG16-based model employing batch normalization, global average pooling, and L2 normalization. This system, designed for corn disease recognition, improved upon the original VGG16 by incorporating transfer learning. Sanida et al. [20] developed a hybrid CNN model that integrates VGG blocks with an inception module for diagnosing tomato crop diseases. Their approach, which leverages data augmentation and a rigorous training process, resulted in outstanding performance metrics. Xu et al. [21] proposed an integrated deep learning algorithm, RFE-CNN, combining residual channel attention blocks, feedback blocks, and elliptic metric learning with a CNN. This model demonstrated superior performance over existing CNNs like VGG-19 and GoogleNet. Khanna et al. [22] introduced PlaNet, a robust CNN model for plant leaf disease recognition. Tested on multiple benchmark datasets, PlaNet exhibited high efficiency and good performance. Attallah [23] proposed RiPa-Net, a pipeline utilizing dual-layers of CNNs along with feature transformation and selection for rice paddy

disease recognition. This approach incorporates spatial-spectral-temporal information. Alkanan and Gulzar [24] enhanced corn seed disease classification through a tailored MobileNetV2 architecture with feature augmentation and transfer learning. Their model, which includes additional layers for improved feature extraction, demonstrated exceptional performance. These studies reflect advancing agricultural diagnosis with deep learning, emphasizing efficiency and accuracy. Therefore, CNN is good at extracting local disease features but lacks the integration of global feature information in crop images. This study introduces the Vision Transformer to overcome this challenge.

B. VIT IN CROP DISEASE DIAGNOSIS

The development of ViT has brought about innovative models in the field of crop disease recognition. Li et al. [25] developed the Interactive Bilinear Transformer Network (IBTN) for fine-grained plant disease recognition, showcasing its effectiveness through public datasets. Zeng et al. [26] introduced the Squeeze-and-Excitation Vision Transformer (SEViT), enhancing classification accuracy in large-scale, fine-grained disease diagnosis by integrating channel attention mechanisms. Tabbakh and Barpanda [27] proposed the TLMViT model, combining transfer learning with Vision Transformers for efficient feature extraction and classification in plant disease diagnosis. Li et al. [28] designed a lightweight vision transformer, PMVT, for real-time plant disease detection on mobile devices, emphasizing computational efficiency and accuracy. Parez et al. [29] presented GreenViT, a finetuned Vision Transformer technique for detecting plant infections and diseases, leveraging the strengths of ViTs to overcome CNN limitations. Fu et al. [30] improved Vision Transformer methods for crop disease image recognition, utilizing block partitioning and self-attention mechanisms for accurate feature extraction. Jin et al. [31] introduced MATSR, a super-resolution method using multiple attention transformers for grape disease recognition, focusing on balancing global and local feature extraction. These innovations reflect the growing application of deep learning in agricultural diagnosis. The previous vision Transformer standard attention mechanism evenly distributes weights on the image without focusing on the lesion area of the crop. Here, we introduce CNN, hybrid-scale, and self-guided attention mechanisms to integrate local and global information. Our attention mechanism can focus on areas with obvious characteristics of crop diseases and exclude interference from different backgrounds.

III. METHODS

A. ARCHITECTURE OF CONVOLUTION SELF-GUIDED TRANSFORMER

The architecture of the original SGT was employed as the foundational structure for the proposed CSGT in this research. The comparison between the two models is shown in Fig.2. The occurrence of Transformer blocks in pairs

is consistently illustrated in Fig.3. In this investigation, we introduce two model variants: base and small. Table 1 outlines the distinctions between each model.

TABLE 1. The detail of architecture variants.

Stage	Size	CSGT(Base)	CSGT(Small)
Stage1	56x56	C1=96,N1=4	C1=64,N1=2
Stage2	28x28	C2=192,N2=6	C2=128,N2=4
Stage3	14x14	C3=384,N4=24	C3=256,N4=16
Stage4	7x7	C4=768,N4=2	C4=512,N4=1

B. CONVOLUTION BLOCK

Our convolution modules are configured to include convolution layers without pooling or activation layers. The calculation for one of the convolution layers is as follows. Considering an image or the input features $X \in \mathbb{R}^{H \times W \times C}$:

$$X_0 = \text{Conv2d}(X) \quad (1)$$

The Conv2d operation employs three filters with a kernel size of 7×7 and padding of 3. This configuration is intended to generate a $224 \times 224 \times 3$ feature map(X_0).

C. HYBRID-SCALE ATTENTION

Illustrated in Fig.4, the input features X undergo projection into Query (Q), Key (K), and Values (V). Subsequently, the multi-head self-attention employs H independent heads. H heads evenly are distributed into h groups, introducing hybrid-scale and multi-receptive field attentions into these h groups, each containing $\frac{H}{h}$ heads. Within the i -th head belonging to the j -th group, with scale s_j (where $j = 1, \dots, h$), every $s_j \times s_j$ tokens in $\{K, V\}$ are consolidated into one token. Following this, we partition $\{Q, K, V\}$ into windows. The window size of $\{K, V\}$ is set to M and remains constant across all groups. To match the window size of $\{Q\}$ with the token merge in $\{K, V\}$, the window size of $\{Q\}$ is selected to be $s_j M \times s_j M$, s_j times larger than that of $\{K, V\}$. This significantly enlarges the receptive field of attention by s_j times:

$$\begin{aligned} Q_i &= XW_i^Q, \\ K_i &= \text{Merge}(X, s_j)W_i^K, \\ V_i &= \text{Merge}(X, s_j)W_i^V, \end{aligned} \quad (2)$$

The function $\text{Merge}(X, s_j)$ represents the consolidation of every $s_j M \times s_j M$ tokens in X into one token, achieved through convolution with a stride of s_j . In the particular scenario where s_j is equal to 1, no token consolidation occurs, and $\{Q, K, V\}$ maintain windows of identical sizes.

$$\begin{aligned} \text{Atten}_i &= \text{Softmax}\left(\frac{P(Q_i, s_j M)P(K_i^T, M)}{\sqrt{d_h}}\right), \\ h_i &= \text{Atten}_i P(V_i, M) \end{aligned} \quad (3)$$

Here, $P(X, s_j M)$ refers to window partitioning with a window size of $s_j M \times s_j M$. Atten_i represents the attention

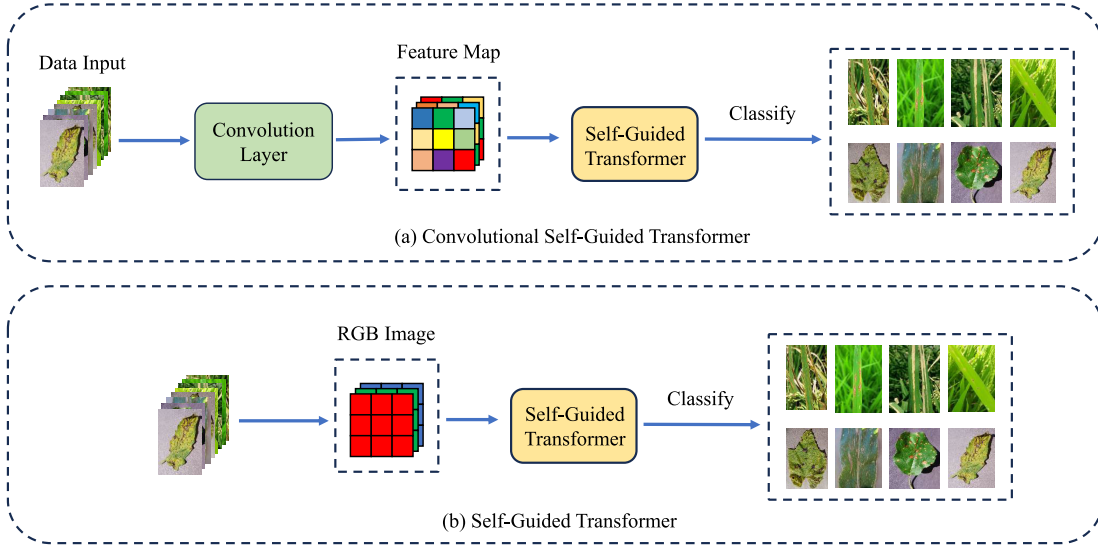


FIGURE 2. Comparison between CSGT and SGT models.

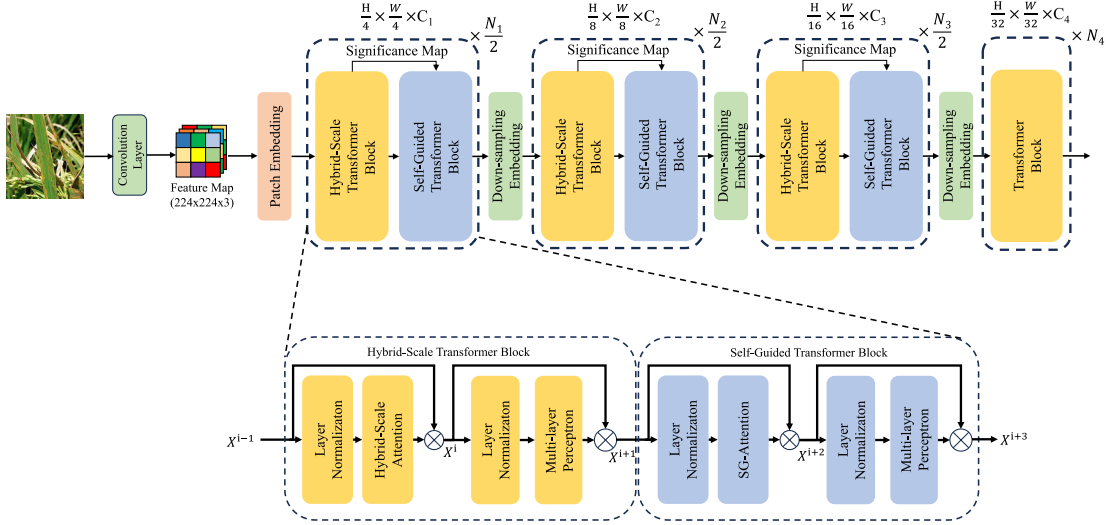


FIGURE 3. Illustration of the convolution self-guided transformer and illustration of two successive transformer blocks.

map. A notable exception occurs when $s_j M \times s_j M$ equals $h \times w$, eliminating the necessity for window partitioning. Consequently, all tokens in $\{K, V\}$ are attended by $\{Q\}$, facilitating global information extraction. The summation of products between all tokens and the current token determines the token's significance:

$$S_i = \frac{1}{hw} \sum_{m=1}^{hw} Atten_i^{m,n},$$

$$S = \sum_{i=1}^h S_i, \quad (4)$$

Here, S represents the final significance map obtained by summing S_i for hybrid-scale guidance, integrating global and fine-grained information.

D. SELF-GUIDED ATTENTION

The input feature map, $X \in \mathbb{R}^{h \times w \times c}$, is initially projected into Query (Q), Key (K), and Value (V). Subsequently, H-independent self-attention heads compute self-attention in parallel. To keep the size of feature maps unchanged after attention, the length of Q is kept fixed while aggregating tokens of K and V using an Importance-guided Aggregation Module (IAM).

$$Q = XW^Q,$$

$$K = \text{IAM}(X, S, r)W^K,$$

$$V = \text{IAM}(X, S, r)W^V. \quad (5)$$

The significance map $S \in \mathbb{R}^{h \times w}$ contains regional significance information across multiple granularities. Sorting the significance map values in ascending order, S is evenly

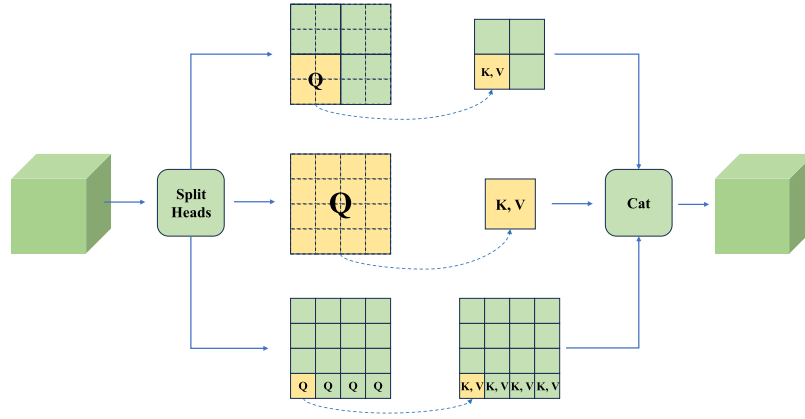


FIGURE 4. Hybrid-Scale attention. self-attention is hybrid-scale and performs attention from local to global areas. it focuses on crop disease areas, captures fine-grained hybrid-scale information, and preserves the global receptive field.

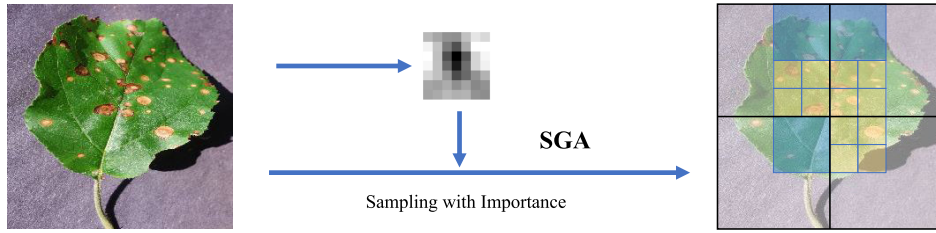


FIGURE 5. Self-Guided Attention. It takes the model-predicted significance maps as guidance for aggregation and preserves more tokens (the blue ones) in salient regions and fewer tokens in minor regions (the yellow ones). The significance depends on the input images and the model.

divided into n sub-regions $S^1, \dots, S^n$. All tokens in X are then categorized into $X^1, \dots, X^n$ based on $S^1, \dots, S^n$. 'r' denotes the aggregation rate where every r tokens are grouped into one token. Different aggregation rates $r_1, \dots, r_n$ are assigned to areas with different significance levels. This approach ensures that regions of higher importance have lower aggregation rates. Consequently, the IAM aggregates grouped input features according to this process (See in Fig.5).

$$\begin{aligned}\hat{X}^i &= F(X^i, r_i), \\ \hat{X} &= \text{Cat}(\hat{X}^1, \dots, \hat{X}^n)\end{aligned}\quad (6)$$

The aggregation function $F(X, r)$ is implemented using fully connected layers, featuring an input dimension of r and an output dimension of 1. The number of tokens in $\hat{X}^i$ corresponds to the number of tokens in X^i divided by r_i .

E. TRANSFORMER BLOCK

Two types of Transformer blocks are correspondingly designed with two self-attention mechanisms, as illustrated in Fig.3. These blocks differ solely at the attention layer, while all other components remain identical:

$$\begin{aligned}\hat{X}^i &= \text{HS-Attention}(\text{LN}(X^{i-1})) + X^{i-1}, \\ \hat{X}^{i+1} &= \text{MLP}(\text{LN}(X^i)) + X^i,\end{aligned}$$

$$\begin{aligned}\hat{X}^{i+2} &= \text{SG-Attention}(\text{LN}(X^{i+1})) + X^{i+1}, \\ \hat{X}^{i+3} &= \text{MLP}(\text{LN}(X^{i+2})) + X^{i+2}\end{aligned}\quad (7)$$

As illustrated in Fig.3, the previous three stages are tailored using hybrid-scale and self-guided Transformer blocks, while for the last stage, we use a vanilla Transformer block following previous Transformers.

IV. RESULTS

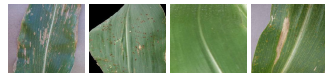
A. DATASETS

1) DATASET1: AGRICULTURAL PEST AND DISEASE DATASET
In the research, we build models using the Agricultural Pest and Disease Dataset [16]. It is a dataset created in 2023. To conduct an in-depth study of crop disease recognition, we selected four crop sub-datasets for separate modeling and analysis (Table 2). The four crops are apples, corn, grapes, and tomatoes from a stable environment. Each image size is $256 \times 256 \times 3$ pixels. The collection contains 20000 photos categorized into 20 crop leaf diseases.

2) DATASET2: RICE LEAF DISEASE DATASET

For the research, we used the Rice Leaf Disease Dataset [17] to examine the model's capacity to distinguish various crop illnesses. This dataset was created in 2023. The images in the Rice Leaf Disease Dataset (Table 2) were

TABLE 2. Information and statistical data of the dataset images.

Class	Name	Number	Data Sample
Apple disease sub-dataset			
00	Black Rot	1000	
01	Cedar Apple Rust	1000	
02	Healthy	1000	
03	Peel Disease	1000	
04	Scab	1000	
Corn disease sub-dataset			
00	Cercospora Leaf Spot	1000	
01	Common Rust	1000	
02	Healthy	1000	
03	Northern Leaf Blight	1000	
Grape disease sub-dataset			
00	Black Measles	1000	
01	Black Rot	1000	
02	Healthy	1000	
03	Leaf Blight	1000	
Tomato disease sub-dataset			
00	Bacterial Spot	1000	
01	Blight	1000	
02	Healthy	1000	
03	Leaf Mold	1000	
04	Mosaic Virus	1000	
05	Septoria Leaf Spot	1000	
06	Yellow Leaf Curl Virus	1000	
Rice Leaf Disease Dataset			
00	Bacterial Blight	238	
01	Bacterial Leaf Streak	162	
02	Brown Spot	753	
03	Healthy	783	
04	Hispa	658	
05	Leaf Blast	634	
06	Sheath Blight	125	

captured in complex environments. The collection contains 3353 photos categorized into 7 rice leaf diseases. This dataset contains different types of image sizes, which are as follows: $300 \times 300 \times 3$ pixels, $187 \times 270 \times 3$ pixels, $1454 \times 1454 \times 3$ pixels, $1515 \times 1515 \times 3$ pixels, $357 \times 270 \times 3$ pixels, and $167 \times 270 \times 3$ pixels.

B. IMPLEMENTATION DETAILS

The model experiments were executed on a high-performance computer (HPC) running the Ubuntu 22.04 LTS operating system. The HPC is configured with four Nvidia RTX4090 GPUs, each possessing 24 GB of memory. The proposed method was implemented using PyTorch 2.0 and Python 3.8. Furthermore, we opted to employ the AdamW optimizer for CSGT to achieve superior training outcomes. The experiment also involved utilizing pre-trained model parameters. After rigorous experimentation, the training protocol entailed 100 epochs, with an initial learning rate set at $1e-3$.

The entire experiment consists of three components: (1) The first component involves image pre-processing, resizing, and data augmentation. (2) The second component focuses on

training the deep learning architecture, which includes setting hyperparameters, input preparation, model training, saving the best model, and selecting the best model for testing. (3) The third component is the assessment of model performance through testing.

C. IMAGE PRE-PROCESSING

Considering that the CSGT network introduced in the research is constructed upon the Self-Guide Transformer, which requires input images of dimensions $224 \times 224 \times 3$, we conducted experimental evaluations using identical image sizes as the inputs of the Self-Guide Transformer network to facilitate a clearer assessment of the enhancements made by the proposed model.

According to the image size specified by the datasets, the original image size provided was inappropriate. Therefore, we employed bilinear interpolation, as described by Bovik [32], to resize the image dimensions. The processed photographs maintained a height and width of 224 pixels without modifying the channels, with the three channels representing red, green, and blue, respectively.

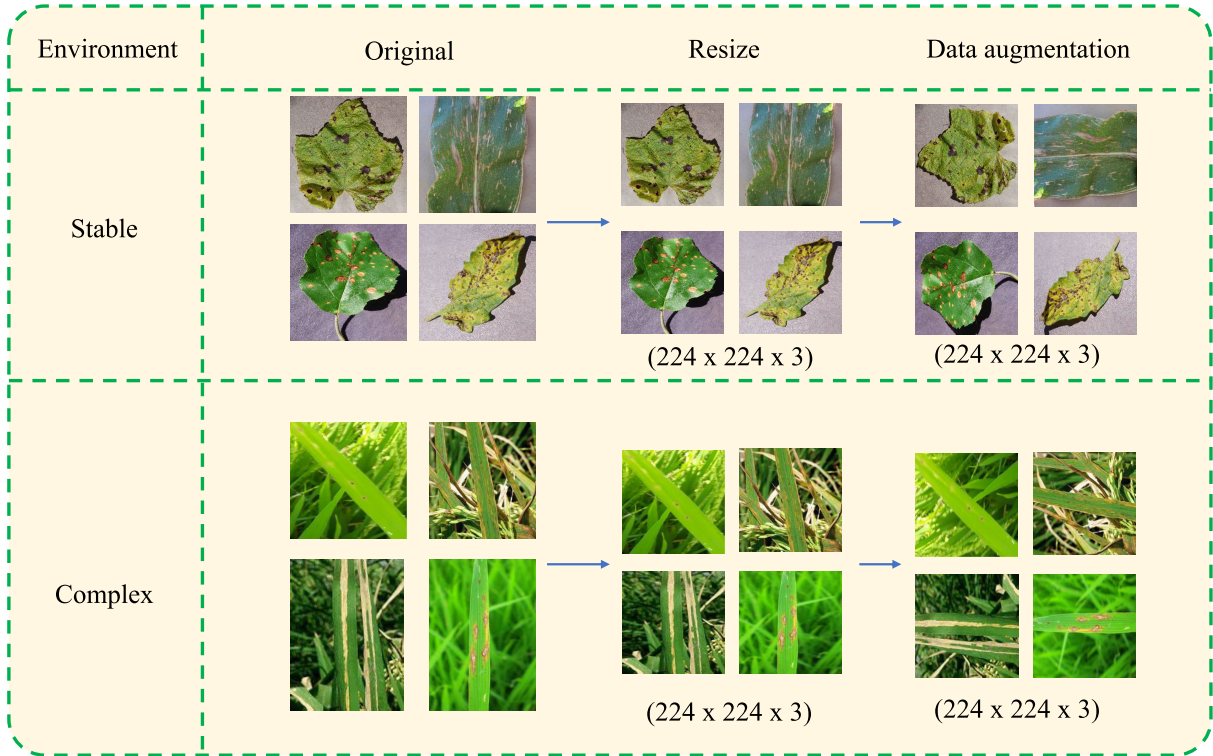


FIGURE 6. Illustration of datasets and data preprocessing.

The quantity of image data available for this study is limited. Training a deep learning model with a small dataset can lead to either overfitting or underfitting. To solve the aforementioned issues and improve the stability and generalizability of the model, we adopted the following data augmentation methods: cropping, flipping, brightness, and rotation. Specific data examples and processes are shown in the figure. Fig.6 displays some examples of augmented images.

D. EVALUATION METRIC

In this paper, the performance of the model was evaluated by criteria such as Accuracy(Acc), precision (P), recall (R), and F1 score. The specific calculation formulas are shown in formulas as follows:

$$\text{Acc} = \frac{TP + TN}{TP + FP + FN + TN} \quad (8)$$

$$P = \frac{TP}{TP + FP} \quad (9)$$

$$R = \frac{TP}{TP + FN} \quad (10)$$

$$\text{F1 Score} = \frac{2 \times P \times R}{P + R} \quad (11)$$

where TP (True Positive) denotes the number of samples correctly classified as the target class, TN (True Negative) represents samples correctly classified as other classes, FP (False Positive) indicates the misclassification

of other-class samples as the target class, and FN (False Negative) signifies misclassification of target-class samples as other classes.

E. CLASSIFICATION ON AGRICULTURAL PESTS AND DISEASES DATASET

The research paper on *Dataset1* does not provide experimental results of the relevant model as a baseline. Therefore, we chose the following common CNN-based and Transformer-based classification models related to the research to test how well our proposed network works adequately. VGG-16 [33], ResNet-50 [34], DenseNet-16 [35], and ConvNeXt [36] were chosen for CNN-based models. Meanwhile, the Transformer-based models selected Swin Transformer [37], Swin Transformer V2 [38], and SG-Former [39].

We evaluated the model's ability to distinguish disease categories using four sub-datasets from the Agricultural Pests and Diseases Dataset in a stable background. We can see from Table 3 that our proposed CSGT has the highest Accuracy and F1 score in four sub-datasets of the Agricultural Pests and Diseases Dataset. The results of each sub-dataset are as follows: Top-1 accuracy reaching 96.9% and F1 score reaching 96.9% in Apple disease. Accuracy reaching 95.8% and F1 score reaching 95.8% in Corn disease. Accuracy reaching 96.1% and F1 score reaching 96.1% in Grape disease. Accuracy reaching 96.5% and F1 score reaching 96.5% in Tomato disease. Compared with the original Self-guided

TABLE 3. Comparison of different models on agricultural pests and diseases.

Models	Params (M)	FLOPs (G)	Acc Apple(%)	F1 score Apple(%)	Acc Corn(%)	F1 score Corn(%)	Acc Grape(%)	F1 score Grape(%)	Acc Tomato(%)	F1 score Tomato(%)
VGG-16	128.06	16.0	89.2	89.3	88.9	88.8	89.3	89.3	89.2	89.2
ResNet-50	22.43	4.0	86.8	86.4	85.4	85.3	86.4	86.5	85.4	85.4
DenseNet-161	25.26	8.0	89.4	89.6	88.5	88.8	87.8	87.7	90.4	90.4
ConvNeXt	82.73	15.5	84.1	83.8	87.5	87.4	89.4	89.5	84.1	84.1
Swin Transformer	82.73	15.4	91.9	91.9	91.5	91.5	91.8	91.8	92.3	92.3
Swin Transformer V2	82.89	20.3	93.1	93.2	92.1	92.1	92.8	92.8	92.6	92.6
SG-Former	73.61	15.6	93.2	93.3	94.3	94.3	93.0	93.1	94.9	94.9
CSGT	73.61	15.6	96.9	96.9	95.8	95.8	96.1	96.1	96.5	96.5

Transformer network, the comparison results of each sub-dataset are as follows: Accuracy increased by 3.7%, and the F1 score increased by 3.6% in the Apple disease. Accuracy increased by 1.5%, and the F1 score increased by 1.5% in the Corn disease. Accuracy increased by 3.1%, and the F1 score increased by 3.0% in the Grape disease. Accuracy increased by 1.6%, and the F1 score increased by 1.6% in the Tomato disease. Our proposed models are the only ones with an accuracy above 95.0% among the models selected in this study. In summary, our proposed models can effectively recognize crop disease species in a stable environment and have higher recognition accuracy than all models included in this study.

F. CLASSIFICATION ON RICE LEAF DISEASES DATASET

The experimental results of related models are provided as a baseline in the research paper of *Dataset2*. To assess the performance of our proposed network, we selected relevant models from the *Dataset2* paper and this study. We adopt Accuracy(Acc), the only experimental result index provided by different models in the paper of *Dataset2* for comparison. Meanwhile, the Transformer-based models selected Swin Transformer [37], Swin Transformer V2 [38], and SG-Former [39].

We will add reference citations as identifiers to the corresponding results. Moreover, the research papers of *Dataset2* do not include model size values, so they are not included in the final comparison results table.

From Table 4, we can note that our proposed network significantly improves the ability of the network to recognize rice leaf diseases compared to other networks. Under this dataset, we can see from Table 4 that CSGT gets the highest Accuracy of 95.8%. Compared with the ECA-ConvNeXt network, Accuracy increased by 1.0% in the Rice disease. Compared with the VGG-19 network, Accuracy increased by 5.2% in the Rice disease. Therefore, our proposed model performs well in classifying different diseases in complex environments.

G. ABLATION EXPERIMENTS AND ANALYSES

See Table 5.

H. QUALITATIVE ANALYSIS

To analyze the effectiveness of our proposed network for classifying crop diseases in different environments, we show

TABLE 4. Comparison of different models on rice leaf.

Model	Acc(%)
VGG-16[17]	93.2
VGG-19[17]	90.6
ResNet-34[17]	93.2
ResNet-50[17]	92.5
ConvNeXt[17]	94.1
ECA-ConvNeXt[17]	94.8
Swin Transformer	92.1
Swin Transformer V2	93.6
SG-Former	94.1
CSGT	95.8

the accuracy of our proposed model in classifying each crop disease in Fig.7 and Fig.9. In the Apple disease images, CSGT obtained an accuracy of 100.0% in classifying the category Peel Disease, as illustrated in Fig.7a. The experiment in the Corn disease sub-dataset showed that the proposed CSGT had a precision of 92.1% or higher for each category, as illustrated in Fig.7b. Considering the disease information in the data set and the recognition rate of the control experiment, the proposed CSGT shows good results. In the Grape disease leaf dataset experiments, the proposed CSGT has a precision of 100.0% for the healthy category, as illustrated in Fig.7c. The accuracy of all other categories was greater than or equal to 91.7%. For experiments in the Tomato disease sub-dataset, CSGT maintained a precision greater than or equal to 91.6% for all categories, as shown in Fig.7d. At the same time, CSGT had a precision of 98.0% or higher in classifying Health, Mosaic Virus, and Yellow Leaf Curl Virus. Overall, CSGT remains effective in classifying the disease category in stable backgrounds. In the Rice Leaf Diseases Dataset experiments, the proposed CSGT had a precision of 98.0% or higher for classifying Bacterial Blight, Bacterial Leaf Streak, and Sheath Blight, as illustrated in Fig.9. Therefore, we believe ours is reliable in classifying multiple classes of diseases in complex environments.

In addition, we also show the confusion matrix in Fig.8 and Fig.10. The confusion matrixes are used to summarize the performance of the proposed classification model. The confusion matrix is normalized over the true condition for our proposed model. Each cell element represents the proportion of the number of the predicted class to the total number of the true class. The diagonal elements represent correctly classified outcomes. All other elements

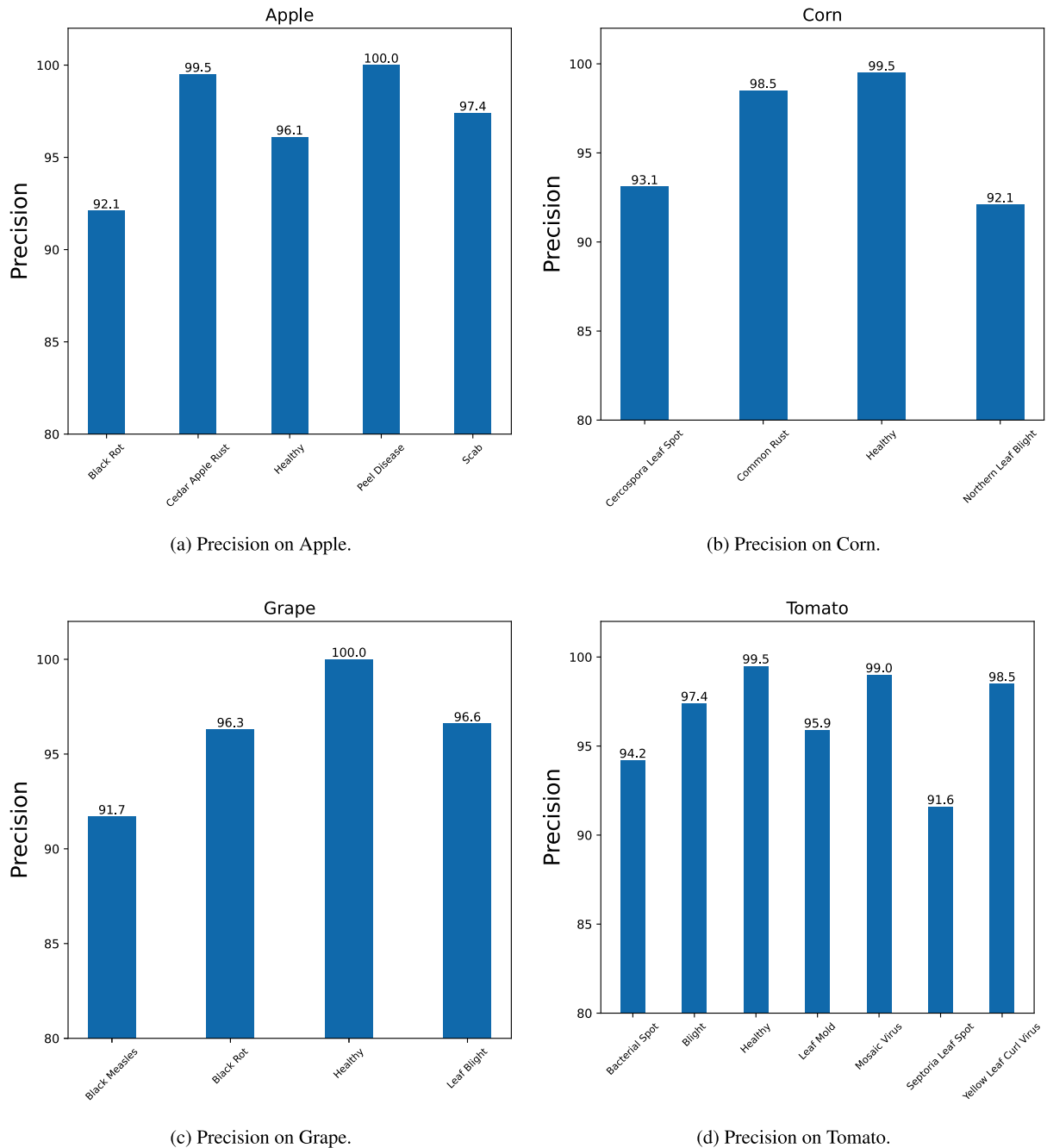


FIGURE 7. Precision of CSGT on agricultural pests and diseases dataset. The precision is presented in percentages (%).

represent incorrectly classified outcomes. There is a one-to-one correspondence between the numerical category labels of the confusion matrices of different datasets and the diseases in the previous dataset description Table 2. By observing Fig.8 and Fig.10, we can see that the proportion of correctly classified images does not differ much, which indicates that our proposed model has no significant classification preference.

The complex environment has an important impact on the model's effective acquisition of crop disease characteristics. This is mainly manifested as a decrease in the mean accuracy

and an increase in the proportion of misclassified images. Compared with image data in a stable environment, data in a real complex environment contains information such as soil, weeds, fingers, and the coexistence of diseased and healthy leaves. The features displayed by this information affect the feature extraction of the model to a certain extent. For example, when multiple leaves are in the image, the model may mainly focus on healthy leaves, resulting in learning failure. In our proposed model, adding convolution layers works well to convert the original images into feature maps to eliminate interference. The main results show that each

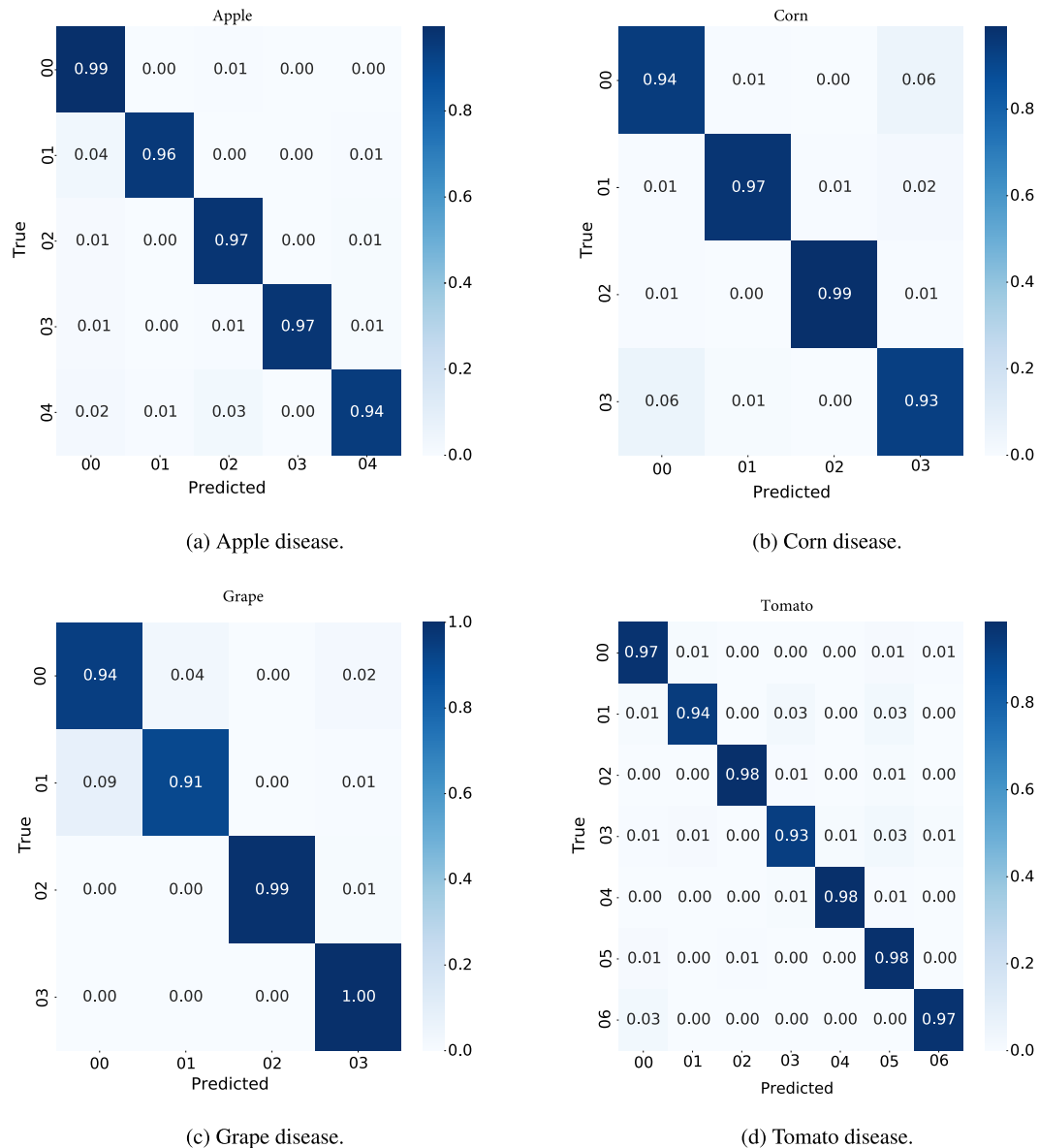


FIGURE 8. Confusion matrixes of CSGT in agricultural pests and diseases dataset.

indicator has improved to varying degrees compared with the original network.

We designed ablation experiments to investigate the effects of pre-trained weight, model size, and CNN layer. The model size in this study incorporated both SG attention and Hybrid-scale attention as an integrated module, with Table 1 outlining the parameter sizes of these components across different models. By conducting ablation experiments with varying model parameters, we effectively investigated the impact of SG and Hybrid-scale attention on the model's performance. The baseline model for this experiment is the base SG-Former network, and the pre-trained model was added. The experimental results of the ablation experiments are shown in Table 5-6. Adding the CNN layer improves the feature extraction capability of the model with

fewer model parameters, resulting in better classification performance.

Ablation experiments in the Agricultural Pests and Diseases Dataset (See in Table 5) showed that the proposed CSGT (Base, 1 CNN layer, With pretraining model) achieved the best precision for each sub-dataset. The comparison results of each sub-dataset are as follows: Top-1 accuracy increased by 3.7% (Apple). Top-1 accuracy increased by 1.5% (Corn). Top-1 accuracy increased by 3.1% (Grape). Top-1 accuracy increased by 1.6% (Tomato). For ablation experiments in the Rice Leaf Diseases Dataset (See in Table 6), Accuracy was improved by 1.7% compared to the baseline model, while F1-Score, Precision, and Recall were also slightly improved by 2.4%, 2.2%, and 1.1%, respectively.

TABLE 5. Ablation experiment on agricultural pests and diseases dataset.

Pre-trained CNN Layer Model Size			Top1-Acc Apple(%)	Top1-Acc Corn(%)	Top1-Acc Grape(%)	Top1-Acc Tomato(%)
Yes	0	Base	93.2	94.3	93.0	94.9
Yes	1	Small	88.2	91.8	91.1	90.9
Yes	3	Small	87.2	87.1	90.6	90.4
Yes	5	Small	87.5	87.9	89.8	88.4
Yes	1	Base	96.9	95.8	96.1	96.5
Yes	3	Base	93.4	91.9	94.3	92.3
Yes	5	Base	91.6	90.9	92.9	90.1
No	1	Small	51.3	62.0	59.8	46.4
No	3	Small	36.4	61.3	55.3	55.8
No	5	Small	48.3	58.6	63.8	42.1
No	1	Base	35.4	50.3	70.8	49.9
No	3	Base	46.3	54.4	69.1	49.9
No	5	Base	39.7	59.9	60.3	43.0

TABLE 6. Ablation experiment on rice leaf diseases dataset.

Pre-trained CNN Layer Model Size			Accuracy(%)	F1-Score(%)	Precision(%)	Recall(%)
Yes	0	Base	94.1	94.8	94.9	94.7
Yes	1	Small	85.8	86.7	86.9	87.0
Yes	3	Small	78.3	78.7	79.7	77.9
Yes	5	Small	77.3	76.0	77.0	75.5
Yes	1	Base	95.8	96.4	97.1	95.8
Yes	3	Base	86.1	87.6	88.8	87.2
Yes	5	Base	79.9	79.3	80.3	78.7
No	1	Small	53.5	48.3	50.9	49.4
No	3	Small	51.1	41.2	40.2	42.4
No	5	Small	46.7	41.4	43.0	41.5
No	1	Base	40.3	24.7	23.0	27.1
No	3	Base	46.1	31.0	31.4	32.0
No	5	Base	36.5	22.3	20.8	24.5

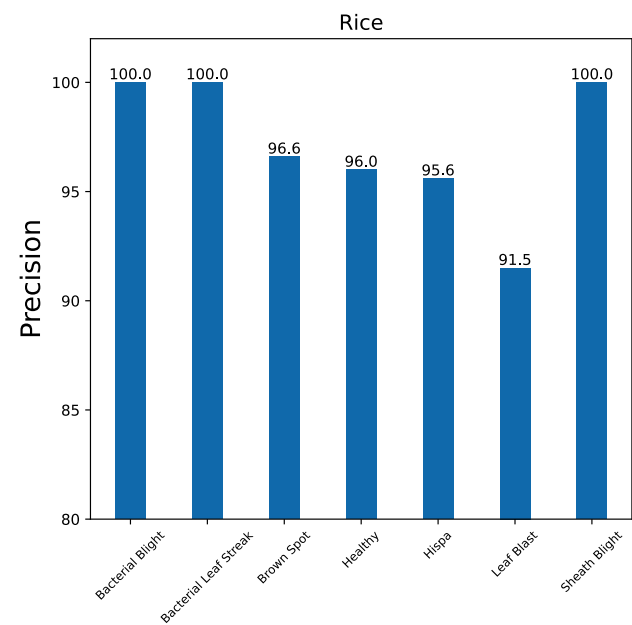


FIGURE 9. Precision of CSGT on rice leaf disease dataset. the precision is presented in percentages (%).

As shown in Table 5-6, for different datasets, the crop disease classification results obtained by the model using



FIGURE 10. Confusion matrix of CSGT in rice leaf disease dataset recognition results.

pre-trained parameters are better than those without using them. When the model uses pre-trained parameters and the

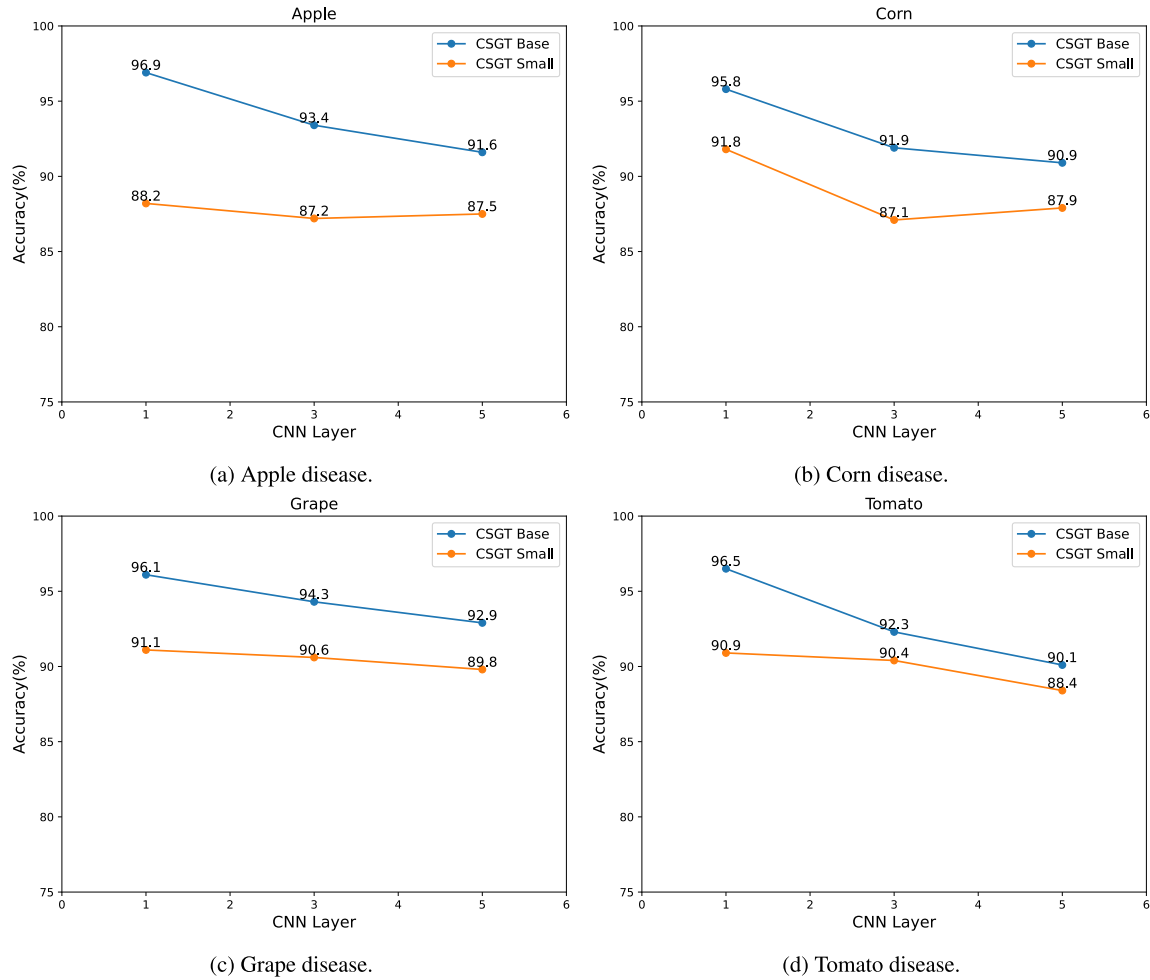


FIGURE 11. Performance of CSGT base and small with varying CNN layers on agricultural pests and diseases dataset.

number of CNN layers is the same, a larger model size will achieve better crop disease classification results than a smaller model size. When the model using pre-trained parameters and the model size are the same, the accuracy of crop disease classification decreases as the number of CNN layers increases(See Fig.11-12).

V. DISCUSSION

First, it is clear from Table 3-4 that the proposed model achieves similar accuracy rates to state-of-the-art models. For the four stable environments sub-datasets in the Agricultural Pests and Diseases Dataset, the highest accuracy and greater than 95.0% was obtained through our proposed CSGT model. Besides, it is worth noting that in the complex environment dataset(the Rice Leaf Disease Dataset), the highest accuracies are achieved by CSGT. Therefore, the performance of our proposed model on the original test set is up to state-of-the-art models.

The inclusion of CNN layers plays a critical role in capturing local features, which is particularly beneficial for identifying distinct disease patterns in crop leaves, thereby

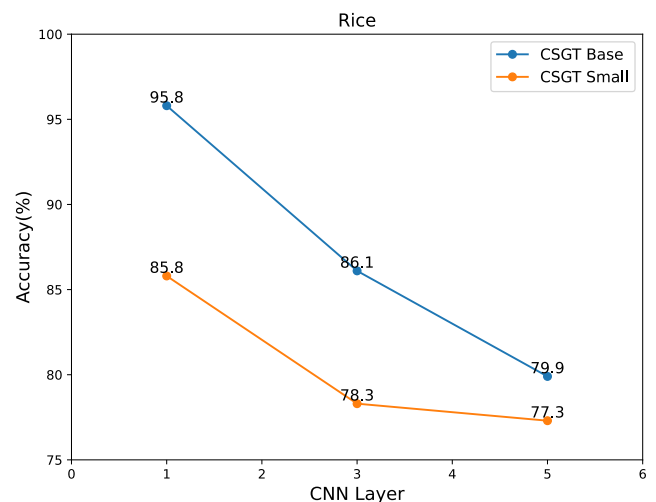


FIGURE 12. Performance of CSGT base and small with varying CNN layers on rice leaf diseases dataset.

contributing to the overall high accuracy of the CSGT model in both stable and complex environments.

Nonetheless, we also explored the influencing factors of the model through ablation experiments. Under the premise of introducing pre-training weights, the number of CNN layers and model sizes will impact the recognition rate of crop diseases. Most models fail to achieve relatively reliable accuracy when the training set is not used with pre-trained weights.

VI. CONCLUSION

This paper proposes the CSGT model based on the Self-guided Transformer to classify crop diseases on two different datasets in different environments. Our model improves the recognition accuracy and training performance by fine-tuning, introducing the convolution layer module, and other effective methods. The accuracy of the model is better than that of other excellent methods. Applying the CSGT model in crop cultivation can potentially decrease yield reductions and economic losses. Based on the similarities of diagnostic methods for diseased leaves of different crops, the proposed model has broad application prospects and can be extended to disease classification of other crops. However, it is important to note that our current agricultural image datasets with complex backgrounds are limited, which presents certain challenges in model generalization. In the future, we will strengthen our collaboration with Zhongkai University of Agriculture and Engineering to expand these datasets and further enhance the platform's performance through multi-modal crop disease diagnosis and model compression.

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